

Project 04: Demand Forecasting for FMCG Products

Time Series Forecasting • Demand Analytics • Python • Business Analytics

Project Overview

This project develops a structured demand forecasting workflow for a mid-sized FMCG company with three core product categories/SKUs: bakery products, beverages, and ready meals. Accurate demand forecasts are essential for improving inventory planning, production scheduling, and supply chain coordination.

This has been implemented by evaluating multiple forecasting approaches using historical weekly sales data and identifying the most suitable model for each product category. The final models were then used to generate forward demand forecasts to support operational planning.

Objective

- Evaluate multiple forecasting approaches using historical demand data
- Compare model performance using walk-forward validation
- Select the most suitable model for each product category
- Generate forward forecasts for operational planning
- Translate forecast outputs into actionable business insights

Tools & Methods

Tools

1. Python
 - a. Pandas
 - b. Statsmodels
2. Excel

Forecasting Models

1. Naïve baseline
2. Rolling mean baseline
3. Exponential Smoothing (ETS)
4. AutoRegressive Integrated Moving Average (ARIMA)

Evaluation Framework

Walk-forward expanding window validation was used to evaluate forecasting performance. Each model generated 12-week forecasts across multiple rolling windows, and performance was measured using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

This approach ensured that models were evaluated on realistic out-of-sample forecasting scenarios.

Key Findings

The model performance varied significantly across product categories. Stable demand series performed well with ETS, while the more volatile beverage demand benefited from ARIMA

Product	Model
Bakery products	ETS
Beverages	ARIMA
Ready meals	ETS

Demand Patterns

- Bakery products: relatively stable with moderate fluctuations, indicating predictable consumption patterns
- Beverages: exhibits higher volatility and a stronger upward trend, suggesting greater sensitivity to demand drivers
- Ready meals: show lower demand levels and a gradual downward trend during the forecast horizon

Deliverables

- [Forecast Dashboard](#)
- [Full Project Repository \(GitHub\)](#)

Insights & Recommendations

- **Production Planning:** Production capacity should be increased for beverages to accommodate the projected growth in demand.
- **Inventory Management:** Bakery products exhibit stable demand, allowing for consistent production and inventory levels.
- **Demand Monitoring:** The downward trend in ready meals should be monitored to avoid overproduction and excess inventory.
- **Forecasting Strategy:** The results highlight that forecasting models should be tailored to product demand characteristics rather than applying a single model across all SKUs.

Limitations

Seasonality was intentionally excluded from the modelling framework to maintain simplicity and focus on core forecasting techniques. Incorporating seasonal components could potentially improve forecast accuracy for products with strong seasonal demand patterns.

Contact

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